Tool Comparison

Giolit SystemVerilog Assertion Verifier 1.0, measured against SymbiYosys and EBMC on public benchmark suites

Summary

On **340 benchmark properties with a known answer** from three public suites, the Giolit SystemVerilog Assertion Verifier (`sva-verify` 1.0) gave **0 wrong answers**. On the EBMC regression suite it decided **272 of 277** such properties. Over all 231 designs with a known answer it decided **219** correctly, where SymbiYosys, the open-source formal flow, decided **104** and EBMC, the open-source model checker whose regression suite this is, decided **160**, with the same time budget per design.

Only `sva-verify` backs every proof with a **certificate**: the inductive invariant, in the design's signal names, signed and checkable by your customer; and only `sva-verify` measures the **safety margin** of every proved bound, with the Giolit Safety Resilience Score.

	sva-verify 1.0	SymbiYosys (open-source flow)	EBMC 6.0
EBMC regression suite: designs decided correctly (of 168 with a known answer)	**163**	**55**	**114**
wrong answers	0	not scored (see "Reading the results")	not scored
verdicts that differ from the published answer, from differences in setup	—	10	4
not supported by the tool's front end or engines	0	103	30
total time for the suite	225 s	40 s	117 s
SymbiYosys examples: designs decided correctly (of 63 with a known answer)	**56**	**49**	**46**
wrong answers	0	not scored (see "Reading the results")	not scored
verdicts that differ from the published answer, from differences in setup	—	3	0

	sva-verify 1.0	SymbiYosys (open-source flow)	EBMC 6.0
not supported by the tool's front end or engines	2	9	12
total time for the suite	181 s	84 s	105 s
TACAS 2016 hardware designs (12 designs, no published answers): designs decided	4	7	6
total time for the suite	142 s	189 s	64 s

What was measured

Benchmarks. Three public suites, used unchanged:

- the **EBMC regression suite** (hw-cbmc project): SystemVerilog and Verilog designs whose expected results (property proved or refuted) are published with each test; tests that only make sense for a bounded check count as having no known answer;
- the **SymbiYosys examples** (YosysHQ): the example designs of the SymbiYosys distribution, with the answers its test files expect;
- the **TACAS 2016** hardware model-checking designs (no published answers; reported as decided or not).

Tools. `sva-verify` 1.0 (trial edition, default options). SymbiYosys from the OSS CAD Suite of 2026-10-05, mode `prove`, engine `abc pdr` (an unbounded IC3/PDR prover), reading each design with Yosys' open-source front end (`read -formal`) and, when that fails, with the yosys-slang plugin. The commercial Verific front end, which extends SymbiYosys' SVA support, was not available and is not part of this comparison; neither are commercial formal tools. **EBMC 6.0** (diffblue/hw-cbmc, the release package), which needs an engine chosen per run: every design gets the same portfolio within the same budget – the default (EBMC's heuristic engine selection), `--k-induction`, `--ic3`, `--bdd`, and `--bound 20` for counterexamples – and per property the first decided answer counts; a bounded "PROVED up to bound" is not counted as a proof. The per-test options in EBMC's own test files were chosen by its developers for each test and are not used.

Settings. One design at a time on the same computer, the same wall-clock budget of 20 s per design for every tool, the same top module, reset and macro definitions.

Scoring, per design. A design is expected to FAIL when one of its properties with a known answer is refuted, otherwise to PASS. `sva-verify` reports each property separately; it decides a design correctly when it finds an expected failure or proves every property with a known answer. SymbiYosys reports one status per design: PASS or FAIL decides it; UNKNOWN or a timeout leaves it open; ERROR means its front end could not read the design. A *wrong answer* is a decided verdict against the published answer.

Reading the results

- **Setup differences are not counted as errors.** On some designs SymbiYosys' verdict differs from the published answer for reasons of setup, not of correctness: the design's only refuted property is a cover (EBMC records a reachable cover as refuted; SymbiYosys' `prove` mode does not check covers), it uses liveness operators its open-source front end does not model, or the test needs an option (a reset, a command-line assertion) that our harness did not pass to it. These designs are listed separately and not scored.
- **Front-end coverage** explains most of the difference: many EBMC tests use SVA sequences, liveness and SystemVerilog constructs that the open-source Yosys flow does not read. `sva-verify` reads them with a full IEEE 1800-2017 elaborator and models the assertions itself.

- **Covers and liveness.** SymbiYosys' `prove` mode does not check cover properties, and the open-source flow rejects most liveness operators; `sva-verify` checks both in the same run.
- **EBMC on its own suite.** The EBMC regression suite was written for EBMC: its published answers are EBMC's own results with options chosen per test. With one uniform portfolio, as a user without that knowledge would run it, EBMC decides fewer of them; several of its engines reject SVA sequences (`unsupported by k-induction`, `not supported by BDD engine`), and some reset settings crash EBMC 6.0. Verdicts that differ from the published answer are listed, not scored.
- **Where SymbiYosys and EBMC are strong.** SymbiYosys' `abc pdr` is a mature bit-level IC3 implementation, and EBMC's SAT-based engines are fast on bit-level designs: on the TACAS 2016 netlists (no published answers) they decide more designs than `sva-verify` within the budget. The per-design results are in the files listed below.
- **Every number here is reproducible**: the verifier package holds the benchmark designs it decides correctly (218 verifications in `benchmarks/MANIFEST.tsv`) and, for each, the console output, the JSON results and a `reproduce.sh` script.

Feature comparison

	sva-verify 1.0	SymbiYosys, open-source flow	EBMC 6.0
Input	SystemVerilog (IEEE 1800-2017) with SVA, via the slang elaborator	Verilog with a subset of SVA (Yosys); more SystemVerilog through the yosys-slang plugin	Verilog and SystemVerilog with SVA (own front end)
Result per property	yes: each assertion PROVED / FAILED / UNDECIDED separately	per task (PASS / FAIL); failing assertions are named	yes, per property
Unbounded proofs	yes (invariants, exhaustive exploration), chosen automatically	yes (`abc pdr`, k-induction engines)	yes (k-induction, IC3, BDD), the engine chosen by the user per run
Proof certificate	**yes: the inductive invariant in the design's signal names, a signed statement with the SHA-256 of every file, a checker for your customer**	no (PASS)	no (PROVED)
Safety margin of proved bounds	**yes: per bound and latency window, with the Giolit Safety Resilience Score (1 to 5 stars)**	no	no
Cover properties	checked in the same run, with witness traces	separate `cover` mode	yes
Liveness (`s_eventually`)	proved by k-liveness or refuted by a lasso	limited in the open-source flow	with chosen engines (liveness-to-safety, ranking functions)
Engines	prover, explicit-state explorer and random simulator raced in parallel, all built in-house	external solvers and engines (ABC, SMT solvers)	SAT-based engines (built-in SAT solvers; SMT solvers optional)

	sva-verify 1.0	SymbiYosys, open-source flow	EBMC 6.0
Counterexamples	per clock cycle, JSON, VCD waveform and a replay testbench for your simulator	VCD waveform files, testbench	per cycle, JSON, VCD
Evidence bundle	certificate or report (PDF), proofs, waveforms, testbenches, `reproduce.sh`	log files	console and JSON
Setup	one command; no project file	a `.sby` project file per task	one command, an engine per run
License	commercial (free trial)	open source (ISC)	open source (BSD)

Measured 13 October 2026 by Giolit Labs. SymbiYosys, Yosys, ABC and EBMC are projects of their respective authors; names are used only to identify the tools measured.